

The fundamental group: generalities

SGA 1, Exposé V

English draft

A. Grothendieck

About this draft

This is an unofficial English translation of Exposé V, “Le groupe fondamental: généralités”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé V. The fundamental group: generalities	1
0. Introduction	1
1. Prescheme with a finite group of operators, quotient prescheme	1
2. Decomposition and inertia groups. Étale case	4
3. Automorphisms and morphisms of étale coverings	7
4. Axiomatic conditions for a Galois theory	8
5. Galois categories	14
6. Exact functors from one Galois category into another	18
7. The case of preschemes	21
8. The case of a normal base prescheme	23
9. The case of non-connected preschemes: multi-Galois categories	23

EXPOSÉ V

The fundamental group: generalities

0. Introduction

The present Seminar is the continuation of the 1960 Seminar. We refer to the latter by symbols such as I 9.7, which means: Séminaire de Géométrie Algébrique, Exposé I, no. 9.7. The numbers of the 1961 exposés will follow those of 1960. We refer to the *Éléments de Géométrie Algébrique* of Dieudonné-Grothendieck by symbols such as (EGA I 8.7.3).

The present exposé summarizes (with slight complements) the last exposés of 1960, which had not been written up.

As in 1961, we shall as a general rule restrict ourselves to locally noetherian preschemes, although this restriction is often inessential. In exposé VI we shall take for granted the theory of faithfully flat descent, summarized in Séminaire Bourbaki no. 190. If appropriate, we shall give a more detailed exposition of it in a later exposé¹, once the reader has had the opportunity to convince himself of the usefulness of this technique for the theory of the fundamental group.

1. Prescheme with a finite group of operators, quotient prescheme

Let X be a prescheme, G a finite group operating on X by automorphisms, on the right to fix ideas. If X is affine with ring A , G therefore operates by automorphisms on the left on A .

For every prescheme Z , G operates on the left on the set $\text{Hom}(X, Z)$, one can therefore consider the set

$$\text{Hom}(X, Z)^G$$

of morphisms invariant under G . It depends functorially on Z , one can ask whether this functor is “representable”, i.e. isomorphic to a functor $Z \mapsto \text{Hom}(Y, Z)$. This means that one can find a prescheme Y , and a morphism invariant under G

$$p: X \longrightarrow Y$$

such that for every Z , the corresponding map $g \mapsto gp$

$$\text{Hom}(Y, Z) \longrightarrow \text{Hom}(X, Z)^G$$

is bijective. One then says that (Y, p) is a *quotient prescheme* of X by G (it is determined up to unique isomorphism).

PROPOSITION 1.1. *Let A be a ring on which the finite group G operates on the left, $B = A^G$ the subring of invariants of A , $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$, $p: X \rightarrow Y$ the canonical morphism (obviously invariant under G). Then*

- (i) *A is integral over B , i.e. p is an integral morphism.*
- (ii) *The morphism p is surjective, its fibers are the orbits of G , the topology of Y is the quotient of that of X .*
- (iii) *Let $x \in X$, $y = p(x)$, G_x the stabilizer of x ; then $k(x)$ is a quasi-Galois algebraic extension of $k(y)$ and the canonical map of G_x into the group $\text{Gal}(k(x)/k(y))$ of $k(y)$ -automorphisms of $k(x)$ is surjective.*
- (iv) *(Y, p) is a quotient prescheme of X by G .*

¹Cf. Exp. VI and Exp. VIII

Statements (i), (ii), (iii) are well known in commutative algebra² and are included only as a reminder, except for the assertion about the topology, which comes from the following general fact, an easy consequence of the Cohen-Seidenberg theorem: an integral morphism is closed (i.e. transforms closed sets into closed sets). Let us note at once:

COROLLARY 1.2. *Under the preceding conditions, the natural homomorphism*

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_X)^G$$

is an isomorphism.

This follows at once from the formula

$$(S^{-1}A)^G = S^{-1}(A^G)$$

valid for every multiplicatively stable subset S of $B = A^G$ (a formula which extends to modules, and which can be stated more generally for a change of base $A \rightarrow A'$ which is *flat*), applied to the case where S is generated by an element f of B .

Assertion (ii) and cor. 1.2 easily imply (iv); more generally, one will have the following:

PROPOSITION 1.3. *Let X be a prescheme with a finite group of automorphisms G , $p: X \rightarrow Y$ an invariant affine morphism such that $\mathcal{O}_Y \xrightarrow{\sim} p_*(\mathcal{O}_X)^G$. Then the conclusions (i), (ii), (iii), (iv) of 1.1 are still valid.*

Indeed, for (i), (ii), (iii), one may assume Y , hence X , affine, and if B, A are their rings, the hypothesis implies $B = A^G$, it suffices to apply 1.1. For (iv), one uses (ii) and $\mathcal{O}_Y = p_*(\mathcal{O}_X)^G$.

COROLLARY 1.4. *Under the conditions of 1.3, for every open set U of Y , U is a quotient of $X|U = p^{-1}(U)$ by G .*

Indeed, $p^{-1}(U) \rightarrow U$ induced by p satisfies the same hypotheses as p .

If now X is a Z -prescheme and the operations of G are Z -automorphisms, then by (iv) Y is a Z -prescheme. That said:

COROLLARY 1.5. *For X to be affine resp. separated over Z , it is necessary and sufficient that Y be so. If X is of finite type over Z , it is finite over Y ; if moreover Z is locally noetherian, Y is of finite type over Z .*

Since X is affine and a fortiori separated over Y , if Y is affine resp. separated over Z , so is X . Conversely, suppose X affine over Z , let us prove that Y is: thanks to 1.4 one may assume Z affine, and one is reduced to proving that if X is affine, Y is, which follows from the explicit determination of Y as $\text{Spec}(A)^G$ made in 1.1. Likewise, since $p: X \rightarrow Y$ is integral, hence universally closed, and surjective, it follows that if X is separated over Z , so is Y (a lemma to be singled out!); indeed, in the diagram

$$\begin{array}{ccc} X \times_Z X & \xrightarrow{p \times_Z p} & Y \times_Z Y \\ \Delta_{X/Z} \uparrow & & \uparrow \Delta_{Y/Z} \\ X & \xrightarrow{p} & Y \end{array}$$

the morphism $X \times_Z X \rightarrow Y \times_Z Y$ is closed, hence transforms the (closed) diagonal of $X \times_Z X$ into a closed subset of $Y \times_Z Y$, which moreover is none other than the diagonal of the latter since p is surjective. — If X is of finite type over Z , it is a fortiori so over Y , hence it is finite over Y (since it is already integral over Y). Suppose moreover Z locally noetherian, let us prove that Y is of finite type over Z . Thanks to 1.4 one may assume Z affine. Since the topological space X is quasi-compact and $p: X \rightarrow Y$ is surjective, Y is likewise quasi-compact, hence a finite union of affine open sets, and by 1.4 one is reduced to the case where Y is affine, hence X affine. But then the ring A of X is an algebra of finite type over the ring C of Z ,

²Cf. N. Bourbaki, Alg. Comm. Chap. 5, §1 and §2, th. 2.

which is noetherian, and it is known that $B = A^G$ is then likewise an algebra of finite type over C (for A will be integral, hence finite over a subalgebra B' of B of finite type over C , hence, since B' is noetherian, B is likewise finite over B' , hence of finite type over C).

COROLLARY 1.6. *For X to be affine resp. a scheme, it is necessary and sufficient that Y be so.*

DEFINITION 1.7. Let X be a prescheme on which a finite group G operates on the right. One says that G *operates admissibly* if there exists a morphism $p: X \rightarrow Y$ having the properties of 1.3 (which implies that X/G exists and is isomorphic to Y).

PROPOSITION 1.8. *Let X be a prescheme on which the finite group G operates on the right. For G to operate admissibly, it is necessary and sufficient that X be a union of affine open sets invariant under G , or again that every orbit of G in X be contained in an affine open set.*

This last condition is obviously implied by the first, and in its turn it implies it; for let T be an orbit of G , U an affine open set containing it; the intersection of the transforms of U by the g in G is then an open set U' stable under G , containing T and contained in the affine open set U . Since in U every finite subset has a fundamental system of affine open neighborhoods, there exists an affine open neighborhood V of T contained in U' . These transforms by the g in G are therefore affine and contained in U' , which is *separated*, hence their intersection U'' is an affine open set which is invariant under G and contains T . — This being established, the condition considered in 1.8 is *necessary*, for one will take the inverse images X_i of affine open sets Y_i covering Y . It is sufficient, for one can then by 1.1 construct the quotients $Y_i = X_i/G$; in each Y_i the image of $X_i \cap X_j$ is an open set Y_{ij} which is identified with X_{ij}/G by 1.4, in particular one deduces from this isomorphisms $Y_{ij} \xrightarrow{\sim} Y_{ji}$ making it possible to glue the Y_i together to construct Y . Serre prefers to construct directly the quotient topological space Y of X by G , to put on it the sheaf $p_*(\mathcal{O}_X)^G$ and to verify that Y becomes a prescheme and that one is then under the conditions of 1.3.

COROLLARY 1.7. *If G operating on X is admissible, the same holds for every subgroup H of G (hence X/H exists).*

This can also be verified directly on the situation 1.3, by noting that one can always assume X affine over some Z and the $s \in G$ operating by Z -automorphisms (one takes for example $Z = Y$); indeed one has:

COROLLARY 1.8. *Suppose X affine over Z , and the operations of G to be Z -automorphisms. Then G operates on X admissibly. If X is defined by a quasi-coherent sheaf $\mathcal{A}$ of algebras, Y is defined by the sheaf $\mathcal{A}^G$ of invariants of $\mathcal{A}$ under G .*

PROPOSITION 1.9. *Suppose that G operates admissibly on X , and that $X/G = Y$ is a prescheme over Z . Consider a change-of-base morphism $Z' \rightarrow Z$, set $X' = X \times_Z Z'$, $Y' = Y \times_Z Z'$, so that G still operates by transport of structure on X' , the morphism $p': X' \rightarrow Y'$ being invariant. If Z' is flat over Z , then p' still satisfies the hypotheses of 1.3, i.e. $\mathcal{O}_{Y'} \rightarrow p'_*(\mathcal{O}_{X'})^G$ is an isomorphism (p' being affine in any case). Hence G operates admissibly on X' , and $(X/G) \times_Z Z' \approx (X \times_Z Z')/G$.*

One may obviously assume $Z = Y$, and one is reduced to the case where moreover Y and Y' are affine. One must show that if B is the subring of invariants of G operating in A , and if B' is an algebra over B which is flat over B , then B' is the subalgebra of invariants of $A' = A \otimes_B B'$. This is immediate, for the exact sequence

$$0 \longrightarrow B \xrightarrow{i} A \xrightarrow{j} A^{(G)}$$

(where the last term means a power of A , and where $j(x)$ is the system of the $s \cdot x - x$, $s \in G$) remains exact under tensoring with B' .

One should take care that the flatness hypothesis was essential for the validity of the result; in particular, if Y' is a closed subscheme of Y (for example even a closed point of Y), X' its inverse image in X , then Y' is *not identified* in general with X'/G . We shall see that it is nevertheless so if X is étale over Y .

To finish, let us give a formalism as convenient as it is trivial. Let Y be a prescheme. Since in the category of preschemes direct sums exist, one can for every set E consider the prescheme sum of a family $(Y_i)_{i \in E}$ of preschemes all identical to Y ; this prescheme will be denoted $Y \times E$. It is characterized by the formula

$$(*) \quad \text{Hom}(Y \times E, Z) = \text{Hom}(E, \text{Hom}(Y, Z))$$

where the second Hom obviously denotes the set of maps from the set E into the set $\text{Hom}(Y, Z)$. One has a canonical morphism

$$Y \times E \longrightarrow Y$$

making $Y \times E$ a prescheme over Y . Since fibered products commute with direct sums (in the category of preschemes) one will have, if Y is a prescheme over some other Z , for a change of base $Z' \rightarrow Z$:

$$(Y \times E) \times_Z Z' = (Y \times_Z Z') \times E$$

(a formula especially useful if $Z = Y$). On the other hand, one concludes trivially from the definition

$$(Y \times E) \times F = Y \times (E \times F) = (Y \times E) \times_Y (Y \times F)$$

(the last formula however resulting from the commutativity pointed out above).

For fixed Y , one can regard $Y \times E$ as a functor in E , with values in preschemes over Y , a functor which commutes with finite products by the preceding formula (which makes it possible for example to associate with every ordinary group G a group scheme $Y \times G$ over Y , which will be finite over Y if Y is, etc.). More generally, this functor is “left exact”, but we shall not need to use this here. This functor also commutes trivially with direct sums, and it is also “right exact”, as one sees at once from the defining formula (*). In particular, if the finite group G operates on the right in the set E , then it operates on the right in $Y \times E$, and one has

$$(Y \times E)/G = Y \times (E/G)$$

where in fact the quotient of the left-hand side satisfies the conditions of 1.3 (this is immediate).

2. Decomposition and inertia groups. Étale case

Let G be a finite group acting on the right on the prescheme X . If $x \in X$, one calls *decomposition group of x* the stabilizer $G_d(x)$ of x . This group acts canonically (on the left) on the residue field $k(x)$, and the set of elements of $G_d(x)$ that act trivially is called the *inertia group of x* , denoted $G_i(x)$.

Suppose that G acts on X in an admissible way and that Y is a prescheme over a prescheme Z . Let us fix a $z \in Z$, and an algebraically closed extension Ω of $k(z)$ having a transcendence degree greater than that of the $k(x)/k(z)$, where x is a point of X over z . One can regard $\text{Spec}(\Omega)$ as a Z -scheme, and the points of X with values in Ω correspond to the homomorphisms of $k(z)$ -algebras $k(x) \rightarrow \Omega$, where x is a point of X over z ; since Ω has been taken large enough, every point x of X over z is the locality of a point of X with values in Ω . If $X(\Omega)$ and $Y(\Omega)$ denote respectively the set of points of X and Y with values in Ω , one has a natural map

$$X(\Omega) \longrightarrow Y(\Omega),$$

on the other hand, G acts on $X(\Omega)$ and the preceding map is invariant under G . This being so, the conclusions (ii) and (iii) of 1.3 can also be interpreted as follows: *the preceding map is surjective and identifies $Y(\Omega)$ with the quotient $X(\Omega)/G$. Moreover, if x is the locality of $a \in X(\Omega)$, then the stabilizer of a in G is none other than the inertia group $G_i(x)$. All this is moreover true without assuming Ω “large enough”; this last hypothesis serves only to ensure*

that one can characterize the inertia group of every element of X over z as a “geometric” stabilizer. One concludes from this, for instance, immediately:

PROPOSITION 2.1. *Let us make an extension of the base $Z' \rightarrow Z$, whence $X' = X \times_Z Z'$. Let x' be a point of X' , x its image in X ; then one has $G_i(x) = G_i(x')$.*

It suffices, in the considerations above, to take for Ω a sufficiently large extension of $k(z')$ (where z, z' are the images of x, x' in Z, Z').

PROPOSITION 2.2. *Under the conditions of 1.3, suppose Y locally noetherian, X finite over Y . Let H be a subgroup of G , consider $X' = X/H$ (cf. 1.7), let $x \in X$, x' its image in X' and y its image in Y .*

- (i) *If $H \supset G_d(x)$, then the homomorphism $\mathcal{O}_y \rightarrow \mathcal{O}_{x'}$ induces an isomorphism on the completions.*
- (ii) *If $H \supset G_i(x)$, then the homomorphism $\mathcal{O}_y \rightarrow \mathcal{O}_{x'}$ is étale, i.e. X' is étale over Y at x' .*

Let $Y_1 = \text{Spec}(\widehat{\mathcal{O}_y})$; let us make the change of base $Y_1 \rightarrow Y$; one finds an $X_1 = X \times_Y Y_1$ finite over Y_1 , on which G acts, the quotient being Y_1 by 1.9. Let y_1 be the unique point of Y_1 over y ; since $k(y) = k(y_1)$, it follows that the fiber of X at y is isomorphic to that of X_1 at y_1 , whence a unique point x_1 of X_1 over x . Moreover, by 1.9 one will have $X_1/H = X'_1 = X' \times_Y Y_1$; let x'_1 be the image of x_1 in X'_1 ; it lies over x' , and one easily checks (X' being of finite type over Y) that the homomorphism $\mathcal{O}_{x'_1} \rightarrow \mathcal{O}_{x'_1}$ induces an isomorphism on the completions. Hence one is reduced to the case where Y is the spectrum of a complete local ring, say B , hence X the spectrum of a ring A finite over B , composed of a finite number of local rings A_x corresponding to the points x_i of X over Y . If A_0 corresponds to $x = x_0$, then A is identified with the ring $\text{Hom}_{G_d}(G, A_0)$ of the functions $f: G \rightarrow A_0$ such that $f(st) = sf(t)$ for $s \in G_d$, the operations of G on these functions being defined by $(uf)(t) = f(tu)$. One sees therefore that if H is an arbitrary subgroup of G , then A^H is the ring of the functions $f: G \rightarrow A_0$ such that

$$f(stu) = sf(t) \quad \text{for } s \in G_d, u \in H$$

hence it is a semi-local ring whose local components correspond to the double cosets $G_d a H$ in G , the double coset defined by $a \in G$ corresponding (thanks to the map $f \mapsto f(a)$) to the subring $A_0^{H(a)}$ of A_0 , where $H(a) = G_d \cap a H a^{-1}$. Moreover, the local component of A^H corresponding to the image x' of x is also the one corresponding to the double coset $G_d H$ of the neutral element; its local component is therefore $A_0^{G_d \cap H}$. If therefore $G_d \subset H$, one finds $A_0^{G_d} = A^G = B$, which proves (i). To prove (ii), one can, by passing to a suitable finite flat extension of A , and using 2.1, reduce to the case where the residue extension $k(x)/k(y)$ is trivial. But then $G_i(x) = G_d(x)$, and one is reduced to the preceding case.

COROLLARY 2.3. *Under the conditions of 2.2, suppose $G_i(x) = (e)$; then X is étale over Y at x . Hence if $G_i(x) = (e)$ for every $x \in X$, then $X \rightarrow Y$ is an étale morphism.*

There is a partial converse:

COROLLARY 2.4. *Suppose X connected and the group G faithful on X . For $p: X \rightarrow Y = X/G$ to be étale, it is necessary and sufficient that the inertia groups of the points of X be reduced to the neutral element. If this is so, G is identified with the group of all Y -automorphisms of the Y -scheme X .*

Taking 2.3 into account, one can suppose X étale over Y . But if an $s \in G$ is in some $G_i(x)$, it then follows from I 5.4 that s acts trivially on G , hence is the unit element since G is faithful, which proves the first assertion. Let u be a Y -automorphism of X , let $x \in X$. By proposition 1.3, there exists an $s \in G$ such that $s(x) = u(x)$, and inducing the same residue homomorphism $k(x)/k(x')$ as u . By loc. cit. one therefore has $s = u$, which completes the proof.

REMARK 2.5. The hypothesis that G acts faithfully is obviously not superfluous in corollary 2.4. The same holds for the hypothesis that X is connected, as one sees for instance by taking $X = Y \times E$, E being a finite set, and G the group of permutations of E : G acts with plenty of inertia, nevertheless $(Y \times E)/G = Y \times (E/G) = Y$, and X is étale over Y . Taking for G a group strictly smaller than the symmetric group of E , but acting transitively on E , one sees that there will also be Y -automorphisms of X not coming from G .

The typical example of a group G acting without inertia is that of $Y \times G$, on which one makes G act by means of its operations on the factor G by right translations: a Y -prescheme X with right group of operators G is said to be *trivial* if it is isomorphic to $Y \times G$.

To make the link between preschemes with finite groups of operators and the notion of principal bundle in a category (a link which, moreover, we shall not need for the rest of the seminar, but which is important in other contexts) the following considerations are useful. We fix a base prescheme Y , and we place ourselves in the category of Y -preschemes. If G is a finite group, we shall set for brevity $G_Y = Y \times G$; it is thus a finite group scheme over Y (cf. no. 1), and if X is a Y -prescheme, one has

$$X \times_Y G_Y = X \times G$$

(same reference). The datum of a Y -morphism $X \times_Y G_Y \rightarrow X$ is therefore equivalent to the datum of a Y -morphism $X \times G \rightarrow X$, i.e. to the datum, for every $g \in G$, of a Y -morphism $T_g: X \rightarrow X$. One notes immediately that for the datum of the T_g to define on X a structure of prescheme with right group of operators G (i.e. $T_{gg'} = T_{g'}T_g$, $T_e = \text{id}_X$) it is necessary and sufficient that the corresponding Y -morphism $X \times_Y G_Y \rightarrow X$ define on X a structure of Y -prescheme with Y -group scheme of operators (in the general sense of objects with $\mathcal{C}$ -group of operators in a category $\mathcal{C}$). Suppose that this is so. Let us recall that X is said to be *formally principal homogeneous* under G_Y ³ if the canonical morphism

$$X \times_Y G_Y \longrightarrow X \times_Y X$$

whose components are respectively pr_1 and the multiplication morphism $\pi: X \times_Y G_Y \rightarrow X$, is an isomorphism. In the present case, identifying the left-hand side with $X \times G$, the morphism considered is the one which, to every $g \in G$, associates the morphism

$$(\text{id}_X, T_g) = (\text{id}_X \times_Y T_g)\Delta_{X/Y}: X \longrightarrow X \times_Y X$$

and consequently, to say that X is formally principal homogeneous under G_Y also means that $X \times_Y X$ is isomorphic to the direct sum of the transforms of the diagonal by the elements (e, g) of $G \times G$ (acting on $X \times_Y X$ in the obvious way), where e denotes the unit element of G . If one wishes not to distinguish left and right, and to give a formula that remains applicable to the product of more than two factors identical to X , one can formulate the condition by saying that the canonical morphism

$$X \times_G (G \times G) \longrightarrow X \times_Y X$$

obtained by attaching to the pair (g, g') the morphism

$$(T_g, T_{g'}) = (T_g \times_Y T_{g'})\Delta_{X/Y}: X \longrightarrow X \times_Y X$$

and by making G act on the left on $G \times G$ via the diagonal homomorphism;

$$s(g, g') = (sg, sg'),$$

is an *isomorphism*.

The notion of *principal homogeneous space* is deduced from that of formally principal homogeneous space by adding a supplementary axiom, ensuring that the “quotient” of X by G_Y exists and is precisely the right unit object of the category, here Y . This axiom may vary according to the context, and is often made explicit most conveniently (in the yoga of “descent”) by requiring that the object with operators become “trivial”, i.e. isomorphic to the

³one now says rather: X is a pseudo-torsor under G_Y .

product $X \times_Y G_Y$ (in the present case $X \times G$) by a suitable change of base, of a specified type (in such a way, in practice, as to permit the technique of descent; cf. Grothendieck, *Technique de descente et théorèmes d'existence en Géométrie Algébrique*, Sémin. Bourbaki no. 190, pages 26 to 28)⁴. In this order of ideas, let us point out here the characterization of principal homogeneous bundles with group G (in the sense of loc. cit.):

PROPOSITION 2.6. *Let Y be a locally noetherian prescheme, X a Y -prescheme with finite group G of operators acting on the right. The following conditions are equivalent:*

- (i) *X is finite over Y , $Y = X/G$, the inertia groups of the points of X are reduced to the unit element.*
- (ii) *There exists a faithfully flat and quasi-compact change of base $Y_1 \rightarrow Y$ such that $X_1 = X \times_Y Y_1$ is a trivial Y_1 -prescheme with operators, i.e. isomorphic to $Y_1 \times G$.*
- (ii bis) *Like (ii), but with $Y_1 \rightarrow Y$ being finite, étale, surjective.*
- (iii) *X is formally principal homogeneous under G_Y , and faithfully flat and quasi-compact over Y .*

Proof (i) $\Rightarrow$ (ii bis) We shall take $Y_1 = X$, noting that $X \rightarrow Y$ is indeed finite, étale by 2.3, and surjective. Let us show that X_1 is then trivial over Y_1 , which will follow from the

COROLLARY 2.7. *If (i) is satisfied and if X admits a section over Y , then X is a trivial space with operators.*

Indeed, this section makes it possible to define a G -morphism $X \times G \rightarrow X$, surjective since G is transitive on the fibers of X , injective since G acts without inertia; finally, it is a local isomorphism by virtue of I 5.3 since X is étale over Y . Hence it is an isomorphism.

(ii bis) trivially implies (ii), which implies (i) because the ingredients of (i) are “invariant” under faithfully flat quasi-compact extension of the base (cf. the Séminaire Bourbaki cited above for “finite”; for the inertia groups, one applies 2.1, and for $Y = X/G$, a converse of 1.9 in the case of a *faithfully flat* change of base, which we had forgotten to make explicit).

We have proved (i) $\Rightarrow$ (iii) in passing, while proving (i) $\Rightarrow$ (ii bis). Finally (iii) $\Rightarrow$ (ii), because the first hypothesis in (iii) means precisely that X becomes trivial on making the change of base $Y_1 = X$; whence (ii) since X is faithfully flat and quasi-compact over Y .

DEFINITION 2.8. A Y -prescheme X with right group of operators G satisfying the equivalent conditions of 2.6 is called a *principal covering of Y , with Galois group G* .

3. Automorphisms and morphisms of étale coverings

PROPOSITION 3.1. *Let X be étale, separated, of finite type over Y , with Y locally noetherian, and let G be a finite group acting on X by Y -automorphisms. Then G acts in an admissible way and the quotient prescheme X/G is étale over Y .*

One does not suppose X finite over Y ; however X is quasi-projective over Y , whence the existence of X/G thanks to 1.8. Let us first prove

COROLLARY 3.2. *The morphism $X \rightarrow X/G$ is étale.*

One can obviously suppose G transitive on the set of connected components of X , then, by considering the stabilizer of a connected component, that X is even connected. Finally, one can suppose that G acts faithfully. But then one sees as in 2.4 that G acts without inertia, hence by 2.3 it follows that $X \rightarrow X/G$ is étale. One concludes thanks to the

LEMMA 3.3 (afterthought to Exposé I). *Let $X \rightarrow X' \rightarrow Y$ be morphisms of finite type, x a point of X , x' and y its images. Suppose Y locally noetherian. If two of the morphisms considered are étale at the marked points, so is the third.*

⁴Cf. Exp. VIII for the theory of flat descent.

It remains only to look at the case where $X \rightarrow X'$ and $X \rightarrow Y$ are étale at x , and to prove that $X' \rightarrow Y$ is étale at x' (which is the case that we need for 3.1). Making a suitable flat extension of the base Y , one is reduced to the case where the residue extension $k(x)/k(y)$ is trivial. Consider the homomorphisms $\mathcal{O}_y \rightarrow \mathcal{O}_{x'} \rightarrow \mathcal{O}_x$ and the homomorphisms deduced by passing to the completions; the hypothesis means that $\widehat{\mathcal{O}}_y \rightarrow \widehat{\mathcal{O}}_x$ and $\widehat{\mathcal{O}}_{x'} \rightarrow \widehat{\mathcal{O}}_x$ are isomorphisms, whence immediately that $\widehat{\mathcal{O}}_y \rightarrow \widehat{\mathcal{O}}_{x'}$ is one, which proves the lemma.

COROLLARY 3.4. *If X is finite and étale over Y , then X/G is finite and étale over Y .*

PROPOSITION 3.5. *Let X, X' be two étale coverings of Y . Then every Y -morphism $f: X \rightarrow X'$ factors as the product of a surjective étale morphism $X \rightarrow X''$ and the canonical immersion $X'' \rightarrow X'$ of a subset X'' of X' that is both open and closed.*

One knows (I 4.8) that f is étale, hence an open morphism; on the other hand, X being finite over Y , f is closed, hence $f(X) = X''$ is a subset of X' that is both open and closed. We are done (N.B. it would have sufficed that X' , instead of being an étale covering, be unramified over Y).

COROLLARY 3.6. *With the preceding notation, $X \rightarrow X'$ is a strict epimorphism in the category of preschemes, and $X' \rightarrow X''$ is a monomorphism (and even a strict monomorphism) in the category of preschemes.*

The first assertion means by definition that the sequence of morphisms

$$X \times_{X''} X \xrightarrow[\text{pr}_2]{\text{pr}_1} X \longrightarrow X''$$

is exact, and this follows from the fact that $X \rightarrow X''$ is finite and faithfully flat, as one sees easily (cf. Grothendieck, loc. cit.). The dual assertion for $X'' \rightarrow X'$ is even more trivial.

Corollary 3.6 will be useful to us for the theory of the fundamental group in the next no.; it is possible (for those who do not like the notion of strict epimorphism) to replace corollary 3.6 by such a variant as the reader will arrange to his personal taste. Let us merely take the opportunity to point out that a factorization $f = f'f''$, with f'' a strict epimorphism and f' a monomorphism, is necessarily unique up to unique isomorphism (in any category); however, there may exist at the same time a factorization $f = f_1f_2$ having the dual properties: f_2 is an epimorphism, f_1 a strict monomorphism (also unique up to unique isomorphism), which is not isomorphic to the preceding one: it suffices to take for instance the category of topological vector spaces (Hausdorff, if one insists), and for $u: X \rightarrow X'$ a morphism such that $u(X)$ is not closed.

PROPOSITION 3.7. *Let Y be a connected locally noetherian prescheme, y a point of Y , Ω an algebraically closed extension of $k(y)$. For every X over Y , one denotes by $X(\Omega)$ the set of points of X with values in Ω . Let X, X' be étale coverings of Y and $u: X \rightarrow X'$ a Y -morphism such that the corresponding map $X(\Omega) \rightarrow X'(\Omega)$ is an isomorphism. Then u is an isomorphism.*

One is immediately reduced to the case where X' is connected. Since $X \rightarrow X'$ is finite and étale, one knows that the geometric number of points in a fiber of $X \rightarrow X'$ is constant, and equal to 1 if and only if the morphism considered is an isomorphism. Now this number is also the number of elements in a fiber of $X(\Omega) \rightarrow X'(\Omega)$, whence the conclusion.

4. Axiomatic conditions for a Galois theory

Let $\mathcal{C}$ be a category, F a covariant functor from $\mathcal{C}$ into the category of finite sets. Suppose the following conditions are satisfied:

- (G 1) $\mathcal{C}$ has a final object⁵ and the fiber product of two objects over a third one in $\mathcal{C}$ exists (this axiom can also be stated by saying that in $\mathcal{C}$ finite projective limits exist).
- (G 2) Finite sums exist in $\mathcal{C}$ (hence so does an initial object $\emptyset_{\mathcal{C}}$, playing the role of the empty set), as does the quotient of an object of $\mathcal{C}$ by a finite group of automorphisms.
- (G 3) Let $u: X \rightarrow Y$ be a morphism in $\mathcal{C}$; then u factors as a product $X \xrightarrow{u'} Y' \xrightarrow{u''} Y$, with u' a *strict* epimorphism and u'' a monomorphism which is an isomorphism onto a direct summand of Y .
- (G 4) The functor F is left exact (i.e. transforms a right unit into a right unit, and commutes with fiber products).
- (G 5) F commutes with finite direct sums, transforms strict epimorphisms into epimorphisms, and commutes with passage to the quotient by a finite group of automorphisms.
- (G 6) Let $u: X \rightarrow Y$ be a morphism in $\mathcal{C}$ such that $F(u)$ is an isomorphism; then u is an isomorphism.

Our object is to construct a topological group π , a projective limit of finite groups, and an equivalence of the category $\mathcal{C}$ with the category $\mathcal{C}(\pi)$ of *finite sets on which π operates continuously* (i.e. in such a way that the stabilizer of a point is an open subgroup, or equivalently that there exists a discrete quotient group which already operates on the set under consideration), the equivalence constructed transforming the given functor F into the obvious inclusion functor of $\mathcal{C}(\pi)$ into the category of finite sets. Let us note right away that the category $\mathcal{C}(\pi)$ constructed by means of a topological group π , and the preceding inclusion functor, do indeed satisfy conditions (G 1) to (G 6).

We proceed in several steps.

- a) Let $u: X \rightarrow Y$ be in $\mathcal{C}$. For u to be a monomorphism, it is necessary and sufficient that $F(u)$ be one. (Uses (G 1), (G 4), (G 6)).

Indeed, to say that u is a monomorphism means that the projection $\text{pr}_1: X \times_Y X \rightarrow X$ is an isomorphism.

- b) Every object X of $\mathcal{C}$ is *artinian*.

Indeed, if $X' \rightarrow X'' \rightarrow X$ are monomorphisms such that $F(X')$ and $F(X'')$ have the same image in $F(X)$, then by a) $F(X') \rightarrow F(X'')$ is an isomorphism, hence $X' \rightarrow X''$ is an isomorphism by (G 6).

- c) The functor F is *strictly pro-representable*. (cf. Grothendieck, Technique de descente et théorèmes d'existence en Géométrie Algébrique, II, Séminaire Bourbaki 195, February 1960).

Indeed, by loc. cit. prop. 3.1, this follows from b) and (G 4). One can therefore find a projective system over a filtered ordered set I :

$$P = (P_i)_{i \in I}$$

in $\mathcal{C}$, considered as a pro-object of $\mathcal{C}$, and a functorial isomorphism

$$(*) \quad F(X) = \text{Hom}_{\text{Pro}(\mathcal{C})}(P, X) = \varprojlim_i \text{Hom}_{\mathcal{C}}(P_i, X)$$

This isomorphism is realized by an element

$$\varphi \in \varprojlim_i F(P_i) = F(P)$$

One may moreover assume that the transition homomorphisms $\varphi_{ji}: P_i \rightarrow P_j$ ($i \geq j$) are *epimorphisms*, and that every epimorphism $P_i \rightarrow P'$ is equivalent to an epimorphism $P_i \rightarrow P_j$ for suitable $j \leq i$ (which determines the projective system P in an essentially unique way).

⁵Recall that an object e of $\mathcal{C}$ is called a *final object* if for every X in $\mathcal{C}$, $\text{Hom}(X, e)$ has exactly one element. One defines dually an *initial object* of $\mathcal{C}$.

An object $X \in \mathcal{C}$ is said to be *connected* if it is not isomorphic to the sum of two objects of $\mathcal{C}$ not isomorphic to the initial object $\emptyset_{\mathcal{C}}$.

d) *The P_i are connected and not isomorphic to $\emptyset_{\mathcal{C}}$.*

If X is a left unit, one has $F(X) = \emptyset$ by (G 5) applied to the direct sum of an empty family, and conversely by (G 6). Hence if X' is an object of $\mathcal{C}$ which is not a left unit, i.e. such that $F(X') \neq \emptyset$, there exists no morphism from X' into X . Hence if some P_i is a left unit, then i is a largest element of the filtered ordered index set I , and formula (*) would mean $F(X) = \text{Hom}(P_i, X) =$ a set reduced to one element for every X , which is absurd since $F(\emptyset_{\mathcal{C}}) = \emptyset$. Hence the P_i are not isomorphic to $\emptyset_{\mathcal{C}}$.

Suppose that one has $P_i = A \amalg B$, whence by (G 5) $F(P_i) = F(A) \amalg F(B)$; in particular the element a_i of $F(P_i)$, corresponding via (*) to the identity homomorphism $P_i \rightarrow P_i$, lies in $F(A) \amalg F(B)$, for instance in $F(A)$. This means that there exists a $j \geq i$ such that $\varphi_{ij}: P_j \rightarrow P_i$ factors as $P_j \rightarrow A \rightarrow P_i = A \amalg B$, where the second arrow is the canonical morphism. Hence $F(P_j) \rightarrow F(P_i)$ factors as $F(P_j) \rightarrow F(A) \rightarrow F(P_i) = F(A) \amalg F(B)$, and since $F(P_j) \rightarrow F(P_i)$ is surjective by (G 5), it follows that $F(B) = \emptyset$, hence B is isomorphic to $\emptyset_{\mathcal{C}}$.

e) *Every morphism $u: X \rightarrow Y$ in $\mathcal{C}$, with X not isomorphic to $\emptyset_{\mathcal{C}}$ and Y connected, is a strict epimorphism. Every endomorphism of a connected object is an automorphism.*

Consider the factorization (G 3) of u ; since $X \neq \emptyset_{\mathcal{C}}$ it follows from (G 6) that $F(X) \neq \emptyset$, hence $F(Y') \neq \emptyset$, hence $Y' \neq \emptyset_{\mathcal{C}}$, hence, Y being connected, Y' is identified with Y , hence u is a strict epimorphism. Suppose that u is an endomorphism of the connected object X ; let us prove that it is an automorphism. Indeed, we may assume X not isomorphic to $\emptyset_{\mathcal{C}}$, hence u is a strict epimorphism by the foregoing, hence $F(u)$ is an epimorphism by (G 5), and since $F(X)$ is a finite set, $F(u)$ is bijective. Hence u is an automorphism by (G 6).

In particular, *every endomorphism of a P_i is an automorphism.*

f) *The following conditions on a P_i are equivalent:* (i) The natural injective map $\text{Hom}(P_i, P_i) \rightarrow \text{Hom}(P, P_i) \simeq F(P_i)$ is also surjective, i.e. for every $u: P \rightarrow P_i$ there exists a $v: P_i \rightarrow P_i$ such that $u = v\varphi_i$ (where φ_i is the canonical homomorphism $P \rightarrow P_i$). (ii) The group $\text{Aut}(P_i)$ operates transitively on $F(P_i)$. (iii) The group $\text{Aut}(P_i)$ operates simply transitively on $F(P_i)$.

Indeed, identifying $\text{Hom}(P, P_i)$ with $F(P_i)$, the map considered in (i) is none other than $v \mapsto F(v)(\varphi_i)$. The equivalence of the three conditions then follows from the fact that $\text{Hom}(P_i, P_i) = \text{Aut}(P_i)$ and that the preceding map is already injective.

A P_i satisfying the equivalent conditions (i), (ii), (iii) of f) is called *Galois*.

g) *For every X in $\mathcal{C}$, there exists a Galois P_i such that every $u \in \text{Hom}(P, X)$ factors as $P \xrightarrow{\varphi_i} P_i \rightarrow X$.*

Let $J = \text{Hom}(P, X) = F(X)$; it is a finite set, hence there exists a P_j such that every $u: P \rightarrow X$ factors as $P \xrightarrow{j} P_j \rightarrow X$, or equivalently such that the natural morphism

$$P \rightarrow X^J \quad (J = \text{Hom}(P, X))$$

factors as

$$P \xrightarrow{\varphi_j} P_j \rightarrow X^J$$

By virtue of (G 3), the morphism $P_j \rightarrow X^J$ factors as the product of a monomorphism and a strict epimorphism, which one may take of the form $\varphi_{ij}: P_j \rightarrow P_i$. We are therefore reduced to proving that P_i is Galois. Let k be an index $\geq j$ such that every morphism $P \rightarrow P_i$ factors through $P \xrightarrow{\varphi_k} P_k \rightarrow P_i$. Note that the natural morphism $P_k \rightarrow X^J$ again factors as the composite

$$P_k \xrightarrow{\varphi_{ik}} P_i \xrightarrow{U} X^J$$

where the first arrow is a strict epimorphism by e), and the second a monomorphism. We want to prove that for a given morphism $\psi: P_k \rightarrow P_i$, there exists an endomorphism v of P_i such that $\psi = v\varphi_{ik}$. But for every $u \in \text{Hom}(P_i, X)$, consider $u\psi \in \text{Hom}(P_k, X)$; it is therefore of the form $u'\varphi_{ik}$ with a well-determined $u' \in \text{Hom}(P_i, X)$. The map $u \mapsto u'$ from J into J thus defined by ψ is moreover injective, since ψ is an epimorphism by virtue of e); it is therefore bijective since the set J is finite. The bijective map $u \mapsto u'$ from J into J therefore defines an isomorphism $\alpha: X^J \xrightarrow{\sim} X^J$ making the diagram

$$\begin{array}{ccccc} P_k & \xrightarrow{\varphi_{ik}} & P_i & \xrightarrow{U} & X^J \\ \parallel & & & \searrow \simeq & \downarrow \alpha \\ P_k & \xrightarrow{\psi} & P_i & \xrightarrow{U} & X^J \end{array}$$

commutative. By the uniqueness properties of the factorization of a morphism as a product of a monomorphism and a strict epimorphism, it follows (since ψ too is a strict epimorphism by e)) that one can find a morphism $v: P_i \rightarrow P_i$ which leaves the diagram commutative, qed.

We conclude in particular that *the Galois P_i form a cofinal system in the system of the (P_j)* . We shall therefore have, since for a Galois object P_i one has

$$\text{Hom}(P, P_i) = \text{Hom}(P_i, P_i) = \text{Aut}(P_i),$$

by passing to the limit:

$$\text{Hom}(P, P) = \varprojlim_i \text{Hom}(P, P_i) = \varprojlim_i \text{Hom}(P_i, P_i) = \varprojlim_i \text{Aut}(P_i)$$

where the projective limit is taken over the Galois P_i . Moreover, by means of the identification $\text{Hom}(P, P_i) = F(P_i)$, and taking into account that F transforms epimorphisms into epimorphisms, one sees that the transition homomorphisms in the preceding projective system are surjective. We conclude from all this:

h) *One has*

$$\text{Hom}(P, P) = \text{Aut}(P) = \varprojlim_i F(P_i) = \varprojlim_i \text{Aut}(P_i),$$

where the projective limit is taken over the Galois P_i .

In particular, $\text{Aut}(P)$ appears as the projective limit of a projective system of finite groups (the transition homomorphisms being surjective); we shall endow it with the projective limit topology of the discrete topologies. *We shall denote by π and call the fundamental group* (of $\mathcal{C}$ equipped with F) *the group opposite to $\text{Aut}(P)$* . This group therefore operates *on the right* on P ; it is the projective limit of finite groups π_i operating on the right on the Galois P_i , π_i being the group opposite to $\text{Aut}(P_i)$.

Taking into account the functorial isomorphism

$$F(X) = \text{Hom}(P, X)$$

and the definition of π , one therefore sees that π operates *on the left* on $F(X)$, and moreover continuously, by g) (for with the notation of g), it is in fact π_i which operates on $F(X)$). It is trivial that for every morphism $u: X \rightarrow Y$ in $\mathcal{C}$, the morphism $F(u): F(X) \rightarrow F(Y)$ is compatible with the operations of π . *One may therefore henceforth regard F as a covariant functor*

$$F: \mathcal{C} \longrightarrow \mathcal{C}(\pi)$$

where $\mathcal{C}(\pi)$ is the category of finite sets on which π operates continuously on the left.

We shall now define a functor in the opposite direction:

$$G: \mathcal{C} \longleftarrow \mathcal{C}(\pi)$$

by the formula

$$G(E) = P \times_{\pi} E,$$

where $P \times_{\pi} E$ is defined as the solution of the universal problem summarized by

$$\mathrm{Hom}_{\mathcal{C}}(P \times_{\pi} E, X) \xrightarrow{\sim} \mathrm{Hom}_{\pi}(E, \mathrm{Hom}(P, X))$$

(where in the right-hand side $\mathrm{Hom}(P, X) = F(X)$ is regarded as a set on which π operates on the left). One must prove the existence of the object $P \times_{\pi} E$.

- i) *Let Q be an object of $\mathcal{C}$ on which a finite group G operates on the right, and E a finite set on which G operates on the left. Then $G \times_G E$ exists, and the canonical map*

$$F(Q) \times_G E \longrightarrow F(Q \times_G E)$$

is an isomorphism.

Since finite direct sums exist in $\mathcal{C}$ by (G 2), and F commutes with them by (G 5), one is immediately reduced to the case where G operates transitively on E , for if the E_j are the orbits of G in E , one will have

$$Q \times_G E = \coprod_j Q \times_G E_j.$$

Let then $a \in E$, and let H be its stabilizer; one sees immediately from the definition that $Q \times_G E$ is identified with Q/H . Whence the existence, thanks to (G 2), and the commutation property for F , thanks to (G 5).

- j) *Let E be an object of $\mathcal{C}(\pi)$, and let P_i be Galois such that π_i already operates on E . Then $P_i \times_{\pi_i} E$ exists and one has a canonical isomorphism*

$$E \xrightarrow{\sim} F(P_i \times_{\pi_i} E)$$

If $j \geq i$ is such that P_j is Galois, then the canonical homomorphism $P_j \times_{\pi_j} E \rightarrow P_i \times_{\pi_i} E$ is an isomorphism.

The first assertion is a special case of i), taking into account that π_i operates simply transitively on $F(P_i)$, which is equipped with a marked point φ_i , whence an isomorphism $F(P_i) \times_{\pi_i} E \simeq E$. For the second assertion one uses for instance (G 6).

For every j , let $\mathcal{C}_j$ be the full subcategory of $\mathcal{C}$ formed by the X such that $\mathrm{Hom}(P_j, X) \rightarrow \mathrm{Hom}(P, X) \simeq F(X)$ is bijective. We know by g) that $\mathcal{C}$ is the filtered union of the $\mathcal{C}_j$. We therefore have, for $X \in \mathcal{C}_j$:

$$\begin{aligned} \mathrm{Hom}_{\pi}(E, \mathrm{Hom}(P, X)) &\simeq \mathrm{Hom}_{\pi}(E, \mathrm{Hom}(P_j, X)) \simeq \mathrm{Hom}_{\pi_j}(E, \mathrm{Hom}(P_j, X)) \\ &\simeq \mathrm{Hom}(P_j \times_{\pi_j} E, X) \end{aligned}$$

and, taking into account the last assertion in j), we find an isomorphism functorial in the object X of $\mathcal{C}_j$:

$$\mathrm{Hom}_{\pi}(E, \mathrm{Hom}(P, X)) \simeq \mathrm{Hom}(P_i \times_{\pi_i} E, X)$$

Since this is true for every j and these functorial isomorphisms, for variable j , induce one another, we conclude:

- k) *Under the conditions of j), the composite of the canonical morphisms*

$$E \longrightarrow \mathrm{Hom}(P_i, P_i \times_{\pi_i} E) \longrightarrow \mathrm{Hom}(P, P_i \times_{\pi_i} E)$$

makes $P_i \times_{\pi_i} E$ a solution of the universal problem defining $P \times_{\pi} E$, i.e. the latter exists and one has an isomorphism

$$P \times_{\pi} E \xrightarrow{\sim} P_i \times_{\pi_i} E$$

This completes the construction of the functor $G(E)$. On the other hand, one has a functorial homomorphism

$$\alpha: \mathrm{id}_{\mathcal{C}(\pi)} \longrightarrow FG$$

i.e. a homomorphism functorial in the object E of $\mathcal{C}(\pi)$:

$$\alpha(E): E \longrightarrow FG(E) = F(P \times_{\pi} E)$$

namely the composite of the canonical morphisms

$$E \longrightarrow F(P) \times_{\pi} E \longrightarrow F(P \times_{\pi} E)$$

(where the first comes from the marked point $\varphi \in F(P)$). Combining j) and k), one finds:

l) *The homomorphism α is an isomorphism*

One defines likewise a functorial homomorphism

$$\beta: GF \longrightarrow \text{id}_{\mathcal{C}}$$

i.e. a homomorphism functorial in the object X of $\mathcal{C}$:

$$\beta(X): P \times_{\pi} F(X) \longrightarrow X$$

as the one associated with the π -homomorphism

$$F(X) \longrightarrow \text{Hom}(P, X)$$

inverse to the canonical isomorphism $\text{Hom}(P, X) \xrightarrow{\sim} F(X)$.

m) *The composites*

$$\begin{aligned} F(X) &\xrightarrow{\alpha(F(X))} FGF(X) \xrightarrow{F(\beta(X))} F(X) \\ G(E) &\xrightarrow{G(\alpha(E))} GFG(E) \xrightarrow{\beta(G(E))} G(E) \end{aligned}$$

are the identity isomorphisms.

A trotting donkey.

Taking l) into account, it follows:

n) *The homomorphism β is an isomorphism*

We have thus obtained the promised result:

THEOREM 4.1. *Let $\mathcal{C}$ be a category satisfying conditions (G 1), (G 2), (G 3) at the beginning of this no., and F a covariant functor from $\mathcal{C}$ into the category of finite sets, satisfying conditions (G 4), (G 5) and (G 6). Then the preceding canonical constructions define equivalences of categories $F: \mathcal{C} \rightarrow \mathcal{C}(\pi)$ and $G: \mathcal{C}(\pi) \rightarrow \mathcal{C}$ quasi-inverse to one another. More precisely, there exists a pro-object P of $\mathcal{C}$, and a functorial isomorphism $F(X) \xleftarrow{\sim} \text{Hom}(P, X)$; π is the group opposite to the group of automorphisms of P , topologized in a suitable way, so that π operates continuously on the sets $\text{Hom}(P, X) \simeq F(X)$. Finally G is given by $G(E) \simeq P \times_{\pi} E$.*

REMARKS 4.2. The statement of conditions (G 1) to (G 6) becomes simpler and more pleasant if one replaces (G 2) and (G 5) respectively by:

(G' 2) Finite inductive limits exist in $\mathcal{C}$.

(G' 5) The functor F is right exact (i.e. commutes with finite inductive limits).

These conditions are apparently stronger than (G 2) and (G 5), but it follows immediately from the structure theorem 4.1 that they are implied by (G 1) to (G 6). Note, however, that in the cases which will interest us, the verification of (G 2) and (G 5) seems in fact simpler than that of (G' 2) and (G' 5). I do not know whether, in condition (G 3), the fact that u'' is an isomorphism onto a direct summand of Y could not be omitted.

5. Galois categories

DEFINITION 5.1. A Galois category is a category $\mathcal{C}$ equivalent to a category $\mathcal{C}(\pi)$, where π is a compact group, projective limit of finite groups (i.e. totally disconnected).

For the definition of $\mathcal{C}(\pi)$, cf. the beginning of no. 4. By Theorem 4.1, $\mathcal{C}$ is Galois if and only if it satisfies conditions (G 1) to (G 3), and if there exists a functor F from $\mathcal{C}$ into the category of finite sets satisfying conditions (G 4) to (G 6) (i.e. which is *exact* and *conservative*, in a general terminology). Such a functor will be called a *fundamental functor* of the Galois category $\mathcal{C}$ ⁶; it is pro-representable by a pro-object which we shall denote by P_F ; a pro-object P such that the associated functor F is fundamental is called a *fundamental pro-object*. In this way, the category of fundamental functors on $\mathcal{C}$ is anti-equivalent to the category of fundamental pro-objects; if F and P correspond to each other, the group $\text{Aut } F$ is therefore isomorphic to the opposite of the group $\text{Aut } P$, hence the group denoted π in the preceding no. is none other than $\text{Aut } P$. Let us recall that in the preceding no. we constructed, starting from a *given* fundamental functor F , an equivalence of $\mathcal{C}$ with $\mathcal{C}(\pi)$ (where $\pi = \text{Aut}(F)$) which transforms the functor F into the canonical functor from $\mathcal{C}(\pi)_x$ into the category of finite sets. In this typical case $\mathcal{C} = \mathcal{C}(\pi)$, $F = \text{canonical functor}$, the fundamental pro-object associated with F is none other than the projective system of the discrete quotients π_i of π .

It may be useful to make explicit the category of pro-objects of $\mathcal{C}(\pi)$. One finds:

PROPOSITION 5.2. *The category $\text{Pro-}\mathcal{C}(\pi)$ is canonically equivalent to the category $\mathcal{C}'(\pi)$ of spaces, with topological group π of operators, which are compact and totally disconnected.*

Since the latter contains $\mathcal{C}(\pi)$ as a full subcategory (corresponding to the compact discrete spaces with operators) and since projective limits exist in it, one has in any case a canonical functor $g: \text{Pro-}\mathcal{C}(\pi) \rightarrow \mathcal{C}'(\pi)$, associating with the projective system $Q = (Q_i)$ the object $X = \varprojlim_i Q_i$ of $\mathcal{C}'(\pi)$. To define a functor in the opposite direction, it suffices to define a contravariant functor from $\mathcal{C}'(\pi)$ into the category of left exact functors $\mathcal{C} \rightarrow \mathbf{Ens}$, and we shall take for every $X \in \mathcal{C}'(\pi)$ the functor $h(X)(E) = \text{Hom}(X, E)$ (the Hom being taken in $\mathcal{C}'(\pi)$). It is immediate by definition that the functors h and g are adjoint to each other, and that hg is canonically isomorphic to the identity functor of $\text{Pro-}\mathcal{C}(\pi)$. It remains to prove (in order to establish that g and h are quasi-inverse to each other) that every object of $\mathcal{C}'(\pi)$ is isomorphic to an object of the form $g(Q)$, where $Q \in \text{Pro-}\mathcal{C}(\pi)$, in other words: *every space X with topological group π of operators, which is compact and totally disconnected, is isomorphic to a projective limit of finite discrete spaces with operators*. Since X is the projective limit of its finite discrete quotients (as a topological space without operators), we are reduced to showing that in the set of these quotients, there is a cofinal system which is invariant under π . For this it suffices to show that for such a quotient X' , the set of transforms of this quotient by the operations of π is finite, (one will then take the sup of the said transforms, which will be an invariant quotient majorizing X'). Or again, that there is an open invariant subgroup π' of π such that the elements of this subgroup leave X' invariant. Now X' corresponds to a finite partition of X into open sets X_i . By reason of continuity and of compactness of π , there exists a neighborhood V of the neutral element of π such that $s \in V$ implies $s \cdot X_i \subset X_i$ for every i , hence s leaves X' invariant. Now we know that the open invariant subgroups of π form a fundamental system of neighborhoods of the neutral element, which completes the proof.

Let us note that one sees even more simply that the category $\text{Ind } \mathcal{C}(\pi)$ is canonically equivalent to the category of sets on which π operates continuously. We shall not need this here.

PROPOSITION 5.3. *Let $\mathcal{C}$ be a Galois category, F a fundamental functor on $\mathcal{C}$, $P = (P_i)$ the associated pro-object, normalized in the usual way. Let $X \in \mathcal{C}$; for X to be connected, it is necessary and sufficient that π operate transitively on $E = F(X)$.*

⁶It seems preferable to adopt the more suggestive term “fiber functor”.

We are reduced to the typical case $\mathcal{C} = \mathcal{C}(\pi)$, $F =$ canonical functor, where it is trivial.

COROLLARY 5.4. *Equivalent conditions on X : (i) X is connected and $\neq \emptyset_{\mathcal{C}}$ (ii) the group π is transitive on $E = F(X)$, and $F(X) \neq \emptyset$ (iii) X is isomorphic to a P_i .*

The equivalence of (i) and (iii) also already follows easily from no. 4, e).

PROPOSITION 5.5. *Let $Q = (Q_i)_{i \in I}$ be a pro-object of $\mathcal{C}$, normalized in the usual way, and let G be the corresponding functor $G(X) = \text{Hom}(Q, X)$ from $\mathcal{C}$ (into **Ens**). The following conditions are equivalent:*

- (i) G commutes with finite direct sums
- (ii) G commutes with the sum of two objects
- (iii) The Q_i are connected and $\neq \emptyset_{\mathcal{C}}$
- (iv) Q is isomorphic to π/H , where H is a closed subgroup of π .
- (v) The functor G is isomorphic to the functor $E \mapsto E^H$ (set of invariants under H) defined by a closed subgroup H of π .

N.B. in the statement of (iv) and (v), one assumes chosen a fundamental functor, making it possible to identify $\mathcal{C}$ with the category $\mathcal{C}(\pi)$.

Proof. We may assume $\mathcal{C} = \mathcal{C}(\pi)$. The implication (i) $\Rightarrow$ (ii) is trivial, (ii) $\Rightarrow$ (iii) is proved like property d) of no. 4. Let us prove (iii) $\Rightarrow$ (iv). Indeed, we know that $\varprojlim_i Q_i$ is nonempty as a projective limit of nonempty finite sets. Let a be a point of $\varprojlim_i Q_i$; it defines a homomorphism of spaces with operators

$$\pi \longrightarrow Q$$

which is *surjective*, since for every i the composite $\pi \rightarrow Q \rightarrow Q_i$ is, because π is transitive on Q_i by virtue of 5.3. If therefore H is the stabilizer subgroup of a , one obtains an isomorphism $\pi/H \xrightarrow{\sim} Q$. The implications (iv) $\Rightarrow$ (v) and (v) $\Rightarrow$ (i) are again trivial.

PROPOSITION 5.6. *Let $\mathcal{C}$ be a Galois category, P a fundamental pro-object of $\mathcal{C}$ and F the associated fundamental functor. Let $P' = (P'_i)_{i \in I}$ be a pro-object of $\mathcal{C}$, put in normal form, and F' the associated functor $F'(X) = \text{Hom}(P', X)$ from $\mathcal{C}$ into **Ens**. Equivalent conditions:*

- (i) $P' \simeq P$, or again $F' \simeq F$.
- (ii) P' is fundamental, or again F' is fundamental.
- (iii) F' transforms a sum of two objects into a sum, and $X \neq \emptyset_{\mathcal{C}}$ implies $F(X) \neq \emptyset$.
- (iv) The objects of $\mathcal{C}$ that are connected and $\neq \emptyset_{\mathcal{C}}$ are exactly the objects isomorphic to a P'_i .

We have trivially (i) $\Rightarrow$ (iii) and (i) $\Rightarrow$ (ii), moreover (ii) $\Rightarrow$ (iv) by virtue of 5.4 (applied to P' instead of P). Moreover (iii) or (iv) implies by virtue of 5.5 that P' is of the form π/H where H is a closed subgroup of π . In case (iii), there exists for every open invariant subgroup π' of π a π -homomorphism $P' = \pi/H \rightarrow \pi/\pi'$, whence $H \subset \pi'$, whence $H = (0)$ and consequently (i), qed.

COROLLARY 5.7. *Let $\mathcal{C}$ be a Galois category. The fundamental pro-objects are isomorphic, the fundamental functors are isomorphic.*

In other words, *the category of fundamental functors is a connected groupoid* Γ , which one may call the *fundamental groupoid* of the Galois category $\mathcal{C}$. If $\mathcal{C} = \mathcal{C}(\pi)$, the group of automorphisms of an object of the fundamental groupoid is isomorphic to π , this isomorphism being well determined up to inner automorphism. (N.B. one calls a *groupoid* a category in which all the morphisms are isomorphisms, a *connected groupoid* a groupoid all of whose objects are isomorphic). The fundamental pro-objects of $\mathcal{C}$ form a connected groupoid equivalent to the *opposite* of the fundamental groupoid. If F, F' are two fundamental functors, associated with fundamental pro-objects P, P' , then $\text{Hom}(F, F') = \text{Isom}(F, F')$ is sometimes denoted $\pi_{F', F}$ and plays the role of a “set of *classes of paths* from F to F' ”. In particular

$\pi_{F,F} = \pi_F$ is none other than the *fundamental group of $\mathcal{C}$* at F constructed in the preceding no. As for the pro-object P associated with F , it plays the role of a *universal covering at F* of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$.

It may be convenient to have a description of $\mathcal{C}$ (up to equivalence) in terms of its fundamental groupoid Γ , without going through a choice of a particular object F of the latter. Now to every object X of $\mathcal{C}$ is associated the functor E_X on the fundamental groupoid, defined by

$$E_X(F) = F(X),$$

with values in **Ens**. (Such a functor is known in topology under the name of “local system” on the groupoid) $F(X) = E_X(F)$ may be called the *fiber* of X at F , and the functor E_X the *fiber-functor* associated with X . The functor E_X has the following property: *for every F , $E_X(F)$ is a finite set on which the topological group $\pi_F = \text{Aut}(F)$ operates continuously*. For a given covariant functor ξ from the fundamental groupoid into **Ens**, the preceding condition is moreover equivalent to the same condition for *one* arbitrary fixed F . This being so:

PROPOSITION 5.8. *The functor $X \mapsto E_X$ is an equivalence of the category $\mathcal{C}$ with the category of covariant functors from the fundamental groupoid Γ of $\mathcal{C}$ into **Ens**, which satisfy the condition highlighted above.*

Indeed, let F_0 be an object of the fundamental groupoid, and let $\pi_0 = \text{Aut}(F_0)$; then the functor $\xi \mapsto \xi(F_0)$ is an equivalence of the second category considered in 5.8 with the category $\mathcal{C}(\pi_0)$, as one sees at once. On the other hand, the composite of the latter and of the functor $X \mapsto E_X$ is the natural equivalence $\mathcal{C} \rightarrow \mathcal{C}(\pi_0)$. It follows that the functor $X \mapsto E_X$ itself is an equivalence.

COROLLARY 5.9. *The category $\text{Pro-}\mathcal{C}$ is canonically equivalent to the category of covariant functors ξ from the fundamental groupoid Γ into the category of topological spaces, satisfying the condition: for every object F of Γ , $\xi(F)$ is a compact totally disconnected space with topological group π_F of operators.*

Here again, one can verify this condition on ξ ; it suffices to verify it for *one* F . The proof is the same as for 5.8.

REMARK 5.10. Let $(F_s)_{s \in S}$ be a family of objects of the fundamental groupoid Γ . Let us set, for $s, s' \in S$:

$$\text{Hom}(s, s') = \text{Hom}(F_s, F_{s'})$$

so that S itself becomes a connected groupoid and the map $s \mapsto F_s$ a fully faithful functor from S into Γ , call it f . Considering then the functor $X \mapsto E_X \circ f$ from $\mathcal{C}$ into the functor category $\text{Hom}(S, \mathbf{Ens})$, one obtains a variant of 5.8 (and 5.9) with Γ replaced by S . The statement thus obtained reduces to Theorem 4.1 when S is reduced to a point, and is none other than 5.8 itself if S is the set of objects of Γ .

We are going to use 5.9 to define a canonical pro-object of $\mathcal{C}$. For this, we consider the functor from Γ into the category of topological spaces (and even of topological groups):

$$f: F \longmapsto \text{Aut}(F) = \pi_F.$$

This functor satisfies the condition considered in 5.8; the space with operators $f(F)$ under π_F is none other than π_F , considered as a space with operators under itself by inner automorphisms. Hence the functor f corresponds to a pro-object of $\mathcal{C}$ determined up to unique isomorphism, which is even a pro-group of $\mathcal{C}$ and which is called the *fundamental pro-group of $\mathcal{C}$* , playing the role of a local system of fundamental groups. It is therefore a pro-group Π of $\mathcal{C}$ defined by the condition that one have an isomorphism functorial in F

$$F(\Pi) \simeq \pi_F.$$

If X is an arbitrary pro-object of $\mathcal{C}$, one has a canonical morphism

$$\Pi \times X \longrightarrow X$$

which makes X an object with a group of operators on the left G in $\text{Pro-}\mathcal{C}$. For this it suffices to note that for F variable, one has a canonical map

$$\Pi(F) \times X(F) \longrightarrow X(F)$$

i.e.

$$\text{Aut}(F) \times E_X(F) \longrightarrow E_X(F), \quad \text{or} \quad \pi_F \times F(X) \longrightarrow F(X)$$

which is functorial in F . It is also functorial in X , hence for every morphism $X \rightarrow Y$ of pro-objects, the corresponding diagram

$$\begin{array}{ccc} \Pi \times X & \longrightarrow & X \\ \downarrow & & \downarrow \\ \Pi \times Y & \longrightarrow & Y \end{array}$$

is commutative.

REMARK 5.11. One must be careful not to confuse a fundamental pro-object P (which is not endowed with a group structure, and is connected) with the fundamental pro-group (which is a pro-group, and in general not connected). Precisely, G is connected if and only if π_F operating on itself by inner automorphisms is transitive, i.e. if π is reduced to the neutral element, or again $\mathcal{C}$ equivalent to the category of finite sets. Another essential difference is that G is determined up to unique isomorphism, and P is determined only up to non-unique isomorphism.

Let E be a finite set and let us consider the constant functor on the groupoid Γ with value E : by virtue of 5.8 it defines an object of $\mathcal{C}$, denoted $E_{\mathcal{C}}$, which may also be interpreted as the sum of E copies of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$. One may consider $E_{\mathcal{C}}$ as a functor in E , from the category of finite sets into the category $\mathcal{C}$, and this functor is *exact*, hence transforms finite groups into $\mathcal{C}$ -groups, etc. If therefore X is an object of $\mathcal{C}$ on which the finite group G operates on the right, we see that one may consider X as an object of $\mathcal{C}$ having a $\mathcal{C}$ -group of operators on the right $G_{\mathcal{C}}$. We shall therefore say, by extension of a general terminology relative to objects with $\mathcal{C}$ -groups of operators, that X is *formally principal homogeneous* under G if X is formally principal homogeneous under $G_{\mathcal{C}}$, i.e. if the canonical morphism

$$X \times G_{\mathcal{C}} \longrightarrow X \times X$$

deduced from the operation of $G_{\mathcal{C}}$ on X on the right, is an isomorphism. We say that X is *principal homogeneous* under G if it is so under $G_{\mathcal{C}}$, i.e. if it is so formally, and if moreover the quotient $X/G = X/G_{\mathcal{C}}$ is $e_{\mathcal{C}}$. If one fixes a fundamental functor, whence an equivalence of $\mathcal{C}$ with a category $\mathcal{C}(\pi)$, X corresponds to a set on which π operates on the left continuously, say $E = F(X)$. Making G operate on X on the right then amounts to making G operate on the set E on the right, in such a way that the operations of G commute with those of π . One then sees at once that X is principal homogeneous under G if and only if the set E is a principal homogeneous space under G , i.e. if and only if G operates on it in a simply transitive way. (Moreover X is formally principal homogeneous if and only if E is principal homogeneous or empty). Comparing with 5.3, we see that if X is principal homogeneous under G and connected, then the given homomorphism from G into the group opposite to $\text{Aut}(X)$ is an *isomorphism*; and moreover, for an object X of $\mathcal{C}$ to be connected and principal homogeneous under the group opposite to $\text{Aut}(X)$, it is necessary and sufficient, with the notation of no. 4, that it be isomorphic to a *Galois* P_i . In the typical case $\mathcal{C} = \mathcal{C}(\pi)$, this means that X is isomorphic to a quotient of π by an invariant subgroup.

Let us still suppose given a fundamental functor F . Then the datum of an X principal homogeneous under a finite group G operating on the right, and of a point $a \in F(X)$, is equivalent to the datum of a homomorphism from π into the group G . Indeed, to such a homomorphism one makes correspond the set $E = G$, making π operate on it on the left by means of the given homomorphism $\pi \rightarrow G$ and the left translations of G , and making G operate on it on the right by right translation, the marked point a of E being the unit

element of G . Thanks to the foregoing, one thus obtains in an essentially unique way every triple (X, G, a) having the properties considered above, since a principal homogeneous set with marked point under a group G is identified with the latter. In this way, one has a direct geometric interpretation of the functor $G \mapsto \text{Hom}(\pi, G)$ from the category of finite groups into **Ens**, a functor which is pro-representable by means of π , and whose consideration would therefore furnish another construction of the group π associated with F .

6. Exact functors from one Galois category into another

PROPOSITION 6.1. *Let $\mathcal{C}, \mathcal{C}'$ be two Galois categories, $H: \mathcal{C} \rightarrow \mathcal{C}'$ a covariant functor, F' a fundamental functor on $\mathcal{C}'$ and $F = F' \circ H$. The following conditions are equivalent:*

- (i) *H is exact, i.e. left exact and right exact.*
- (ii) *H is left exact, transforms finite sums into finite sums, and epimorphisms into epimorphisms (or again: objects $\not\approx \emptyset_{\mathcal{C}}$ into objects $\not\approx \emptyset_{\mathcal{C}'}$).*
- (iii) *F is a fundamental functor on $\mathcal{C}$.*

The implication (i) $\Rightarrow$ (ii) is a general fact about categories. Moreover the first form given to (ii) implies the second, as one sees by noting that if X is an object of $\mathcal{C}$, then X is $\not\approx \emptyset_{\mathcal{C}}$ if and only if the morphism $X \rightarrow e_{\mathcal{C}}$ is an epimorphism; one will note that F , being assumed left exact, transforms $e_{\mathcal{C}}$ into $e_{\mathcal{C}'}$. The second form given to (ii) implies (iii), since F , being left exact and hence pro-representable, falls under 5.6 criterion (iii). Finally (iii) implies (i), as follows from the fact that F is exact, and “conservative” (i.e. satisfies axiom (G 6) of no. 4).

Let then Γ be the fundamental groupoid of $\mathcal{C}$, and Γ' that of $\mathcal{C}'$. Thus if H is exact, then $F' \mapsto F' \circ H$ is a functor from the groupoid Γ' into the groupoid Γ , which we shall denote by tH :

$${}^tH(F')(X) = F'(H(X))$$

which can also be written, with the notation $F(X) = E_X(F)$ introduced in no. 6:

$$E_{H(X)}(F') = E_X({}^tH(F')).$$

This last formula shows, taking into account 5.8 or 4.1, that the exact functor H is determined (up to unique isomorphism) when one knows the corresponding functor tH . Let us fix an F' , and let $F = {}^tH(F')$; then tH defines a homomorphism from $\pi_{F'} = \text{Aut}(F')$ into $\pi_F = \text{Aut}(F)$:

$${}^tH: \pi_{F'} \longrightarrow \pi_F \quad (F = {}^tH(F') = F' \circ H).$$

Moreover the formula above shows (taking into account 5.8) that this homomorphism has the following property: for every finite set E on which π_F operates continuously, the group $\pi_{F'}$ also operates *continuously* by means of the preceding homomorphism $\pi_{F'} \rightarrow \pi_F$. Applying this to the quotients of π_F by its open invariant subgroups, one sees that the preceding condition also means that the homomorphism under consideration is continuous. Conversely, let us be given an object F of Γ , an object F' of Γ' and a continuous homomorphism

$$u: \pi_{F'} \longrightarrow \pi_F,$$

to it there therefore corresponds a functor from $\mathcal{C}(\pi)$ into $\mathcal{C}(\pi')$, manifestly exact, hence by virtue of 4.1 there corresponds to it a functor H from $\mathcal{C}$ into $\mathcal{C}'$ which is exact, and such that ${}^tH: \pi_{F'} \rightarrow \pi_F$ is precisely u . One can also, instead of a homomorphism of groups, start from a *functor*

$$U: \Gamma' \longrightarrow \Gamma$$

which is such that for *every* $F' \in \Gamma'$ (or for *one* $F' \in \Gamma'$, which amounts to the same) the corresponding homomorphism $\pi_{F'} \rightarrow \pi_F$ is continuous: such a functor is isomorphic to a functor of the form tH , where $H: \mathcal{C} \rightarrow \mathcal{C}'$ is an exact functor determined up to unique isomorphism. Thus:

COROLLARY 6.2. *For a functor $H: \mathcal{C} \rightarrow \mathcal{C}'$ between Galois categories to be exact, it is necessary and sufficient that there exist equivalences $\mathcal{C}(\pi) \rightarrow \mathcal{C}$ and $\mathcal{C}' \rightarrow \mathcal{C}(\pi')$ which transform the functor H into the functor $\mathcal{C}(\pi) \rightarrow \mathcal{C}(\pi')$ associated with a homomorphism of topological groups $\pi' \rightarrow \pi$.*

COROLLARY 6.3. *Let $\mathcal{C}, \mathcal{C}'$ be two Galois categories, and Γ, Γ' their fundamental groupoids. Then the category of exact functors from $\mathcal{C}$ into $\mathcal{C}'$ is equivalent to the category of functors $U: \Gamma' \rightarrow \Gamma$ having the following property: for every F' in Γ' (or for one F' in Γ' , which amounts to the same), setting $F = U(F')$, the homomorphism*

$$\pi_{F'} = \text{Aut}(F') \longrightarrow \pi_F = \text{Aut}(F)$$

defined by U is continuous.

Consider the fundamental pro-group Π of $\mathcal{C}$; then an exact functor H transforms it into a pro-group $H(\Pi)$ of $\mathcal{C}'$; we are going to define a homomorphism

$$\Pi' \longrightarrow H(\Pi)$$

(where Π' is the fundamental pro-group of $\mathcal{C}'$), by the condition that for every object F' of Γ' , the corresponding homomorphism

$$F'(\Pi') = \pi_{F'} \longrightarrow F'(H(\Pi)) = \pi_F \quad (\text{where } F = F' \circ H = {}^t H(F'))$$

be the natural homomorphism

$$\text{Aut}(F') \longrightarrow \text{Aut}(F' \circ H).$$

Since the latter is functorial in F' , it indeed defines, by virtue of 5.8, a homomorphism of pro-objects, and in fact of pro-groups, of $\mathcal{C}'$. This homomorphism is said to be *associated* with the functor H .

Now let H' be a second exact functor, from the Galois category $\mathcal{C}'$ into a Galois category $\mathcal{C}''$. It is trivial that one has

$${}^t(H'H) = {}^t H {}^t H'$$

(N.B. we have here an identity of functors, and not merely a canonical isomorphism). One has an analogous transitivity property for the associated homomorphisms of the fundamental pro-groups.

We shall now interpret the properties of the exact functor H in terms of the corresponding homomorphism

$$u: \pi_{F'} \longrightarrow \pi_F \quad (\text{where } F' = F' \circ H).$$

It is convenient to introduce the notion of a *pointed object* of the Galois category $\mathcal{C}$ (endowed with its fundamental functor F): it is by definition an object X of $\mathcal{C}$ endowed with an element a of $F(X)$. It is therefore interpreted as a finite set on which π_F operates continuously on the left, endowed with a point a . Hence the *connected* pointed objects of $\mathcal{C}$ are identified, by virtue of 5.3, with the open subgroups of π_F . If U and V are two such subgroups, corresponding to connected pointed objects X, Y of $\mathcal{C}$, then there exists a pointed homomorphism from X into Y if and only if $U \subset V$, and this homomorphism is then unique. Of course, the functor H transforms pointed objects into pointed objects (since $F = F' \circ H$). Let us note on the other hand that a closed subgroup of a group such as π_F is the intersection of the open subgroups that contain it; consequently, if M, N are two closed subgroups, then $M \subset N$ if and only if every open subgroup that contains N also contains M . Thanks to these remarks, one easily proves the results that follow:

PROPOSITION 6.4. *Let X be a connected pointed object of $\mathcal{C}$, associated with an open subgroup U of π_F . In order that U contain $u(\pi_{F'})$ (resp. the closed invariant subgroup generated by $u(\pi_{F'})$), it is necessary and sufficient that $H(X)$ admit a pointed section (resp. be completely decomposed).*

We shall call a *section* — understood: over the final object — of an object X of a Galois category $\mathcal{C}$ a morphism from the final object $e_{\mathcal{C}}$ into X , which amounts to giving an element a of $F(X)$ invariant under π_F ; if X is pointed, one says that one has a *pointed section* if it is compatible with the pointed structures on X and $e_{\mathcal{C}}$, i.e. if a is precisely the marked object of $F(X)$. Such a section is therefore unique, and exists if and only if the marked object of $F(X)$ is invariant under π_F . Finally, an object of a Galois category is said to be *completely decomposed* if it is isomorphic to a sum of final objects, i.e. if π_F operates trivially on $F(X)$ — a condition evidently stronger than the existence of a pointed section, when X is pointed. Proposition 6.4 follows trivially from the preceding definitions and remarks.

COROLLARY 6.5. *For u to be trivial, it is necessary and sufficient that for every object X of $\mathcal{C}$, $H(X)$ be completely decomposed.*

PROPOSITION 6.6. *Let X' be a connected pointed object of $\mathcal{C}'$, associated with an open subgroup U' of $\pi_{F'}$. In order that U' contain $\text{Ker } u$, it is necessary and sufficient that there exist a connected pointed object X of $\mathcal{C}$ and a pointed homomorphism from the pointed connected component X'_0 of $H(X)$ into X' (hence that X be isomorphic as a pointed object to a quotient of the neutral connected component of the inverse image of a pointed object of $\mathcal{C}$). If u is surjective, the preceding condition is also equivalent to the following: X is isomorphic to an $H(X)$, where X is a pointed object of $\mathcal{C}$.*

(One calls *neutral connected component* of a pointed object X of a Galois category $\mathcal{C}$ the unique connected pointed subobject of X ; it corresponds to the orbit under π_F of the marked point of $F(X)$, by virtue of 5.3). Since the fact that U' contains $\text{Ker } u$ does not depend on the chosen pointing of X' (because another pointing amounts to replacing U by a subgroup conjugate to U), one sees:

COROLLARY 6.7. *In order that U' contain $\text{Ker } u$, it is necessary and sufficient that there exist an object X of $\mathcal{C}$ (which one may assume connected) and a morphism from a connected component of $H(X)$ into X' . If u is surjective, this also means that X' is isomorphic to an object of the form $H(X)$.*

COROLLARY 6.8. *In order that u be injective, it is necessary and sufficient that for every object X' of $\mathcal{C}'$ there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ into X' .*

PROPOSITION 6.9. *The following conditions are equivalent:*

- (i) *The homomorphism $u: \pi_{F'} \rightarrow \pi_F$ is surjective.*
- (ii) *For every connected object X of $\mathcal{C}$, $H(X)$ is connected.*
- (iii) *The functor H is fully faithful.*

This last fact means that for two objects X, Y of $\mathcal{C}$, the natural map

$$\text{Hom}(X, Y) \longrightarrow \text{Hom}(H(X), H(Y))$$

is bijective.

COROLLARY 6.10. *For u to be an isomorphism, it is necessary and sufficient that H be an equivalence of categories, or again that the following two conditions be satisfied:*

- a) *for every connected object X of $\mathcal{C}$, $H(X)$ is connected*
- b) *every object of $\mathcal{C}'$ is isomorphic to an object of the form $H(X)$.*

PROPOSITION 6.11. *Let $H: \mathcal{C} \rightarrow \mathcal{C}'$ and $H': \mathcal{C} \rightarrow \mathcal{C}''$ be exact functors between Galois categories, and F'' a fundamental functor on $\mathcal{C}''$; let us set $F' = F''H'$ and $F = F'H$, and consider the associated homomorphisms*

$$u': \pi_{F''} \longrightarrow \pi_{F'} \quad u: \pi_{F'} \longrightarrow \pi_F.$$

In order that $\text{Ker } u \subset \text{Im } u'$, i.e. in order that uu' be the trivial homomorphism, it is necessary and sufficient that for every object X of $\mathcal{C}$, $H'(H(X))$ be completely decomposed. In order that

$\text{Ker } u \supset \text{Im } u'$, it is necessary and sufficient that for every connected pointed object X' of $\mathcal{C}'$ such that $H'(X')$ admits a pointed section, there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ into X' .

The first assertion follows from the last statement of 6.4. The second follows from the conjunction of 6.4 and 6.6.

REMARK 6.12. It is not true in general, under the conditions of 6.8, that X' is isomorphic to an object of the form $H(X)$. One can show that in order that every connected object (hence every object) of $\mathcal{C}'$ be isomorphic to an object of the form $H(X)$, it is necessary and sufficient that u be an isomorphism of $\pi_{F'}$ onto a subgroup of π_F that is a *direct factor*. In practice, however, one directly constructs a homomorphism $\pi_F \rightarrow \pi_{F'}$ right inverse to u by means of a suitable exact functor from $\mathcal{C}'$ into $\mathcal{C}$.

PROPOSITION 6.13. *Let $\mathcal{C}$ be a Galois category endowed with a fundamental functor F , S a connected object of $\mathcal{C}$, and $\mathcal{C}'$ the category of objects of $\mathcal{C}$ over S . Then $\mathcal{C}'$ is a Galois category, and the functor $X \mapsto H(X) = X \times S$ from $\mathcal{C}$ into $\mathcal{C}'$ is exact. Let $a \in F(S)$, and let F' be the functor from $\mathcal{C}'$ into the category of finite sets defined by*

$$F'(X') = \text{inverse image of } a \text{ under } F(X') \longrightarrow F(S).$$

Then one has an isomorphism $F \cong F' \circ H$, and the corresponding homomorphism

$$u : \pi_{F'} \longrightarrow \pi_F$$

is an isomorphism of $\pi_{F'}$ onto the open subgroup U of π_F that is the stabilizer of the marked element a of $F(X)$.

The proof is left to the reader.

7. The case of preschemes

Let S be a locally noetherian and *connected* prescheme, and let

$$a : \text{Spec}(\Omega) \longrightarrow S$$

be a geometric point of S , with values in an algebraically closed field Ω . We shall set

$$\mathcal{C} = \text{category of étale coverings of } S$$

and for an object X of $\mathcal{C}$, i.e. an étale covering X of S , we set

$$F(X) = \text{set of geometric points of } X \text{ over } a.$$

Thus F becomes a functor on $\mathcal{C}$ with values in the category of finite sets. Properties (G 1) to (G 6) are satisfied: for (G 1), this is contained in the sorites of I 4.6, (G 2) follows from 3.4, (G 3) from 3.5, (G 4) is trivial by definition, (G 5) follows from 3.5 and from the beginning of no. 2, finally (G 6) is proved in 3.7. One can therefore apply the results of no. 4, 5, 6. This makes it possible in particular to define a pro-object P of $\mathcal{C}$ representing F , called the *universal covering of S at the point a* , and a topological group $\pi = \text{Aut}(F) = \text{Aut}^0 P$, called the *fundamental group of S at a* , and denoted $\pi_1(S, a)$. The functor F then defines an equivalence of the category $\mathcal{C}$ with the category of finite sets on which $\pi = \pi_1(S, a)$ operates continuously. This equivalence therefore makes it possible to interpret the usual operations of projective limits and finite inductive limits on coverings (products, fiber products, sums, passage to the quotient, etc.) in terms of the analogous operations in $\mathcal{C}(\pi)$, i.e. in terms of the obvious operations on finite sets on which π operates. Let us note moreover, since the topological connected components of an étale covering are also étale coverings, that *an object X of $\mathcal{C}$ is connected in $\mathcal{C}$ if and only if it is topologically connected*; by virtue of 5.3 this therefore means that π_1 operates transitively on $F(X)$. Let us note that in order for an object X of $\mathcal{C}$ to be faithfully flat and quasi-compact over S (as it is already flat and quasi-compact over S), it is necessary and sufficient that $X \rightarrow S$ be surjective, i.e. be an epimorphism in $\mathcal{C}$, or again that $X \neq \emptyset$. One then concludes from criterion 2.6 (iii) that X

is a principal covering of S with group G if and only if it is a principal homogeneous space under G in the category $\mathcal{C}$, (as it was defined in no. 5).

If a' is another geometric point of S (corresponding to an algebraically closed field Ω' , which may be different from Ω and which may even have a different characteristic), it defines a fiber functor $F' = F_{a'}$ from $\mathcal{C}$ into the category of finite sets, which is again exact, hence isomorphic to $F = F_a$. Consequently, the fundamental groups $\pi_1(S; a)$ for variable a are isomorphic to one another. If one denotes by $\pi_1(S; a, a')$ the set of isomorphisms (or, what amounts to the same, the set of homomorphisms) $F_a \rightarrow F_{a'}$ of the associated fiber functors, one thus obtains a *groupoid* whose set of objects is the set of geometric points of S , the fundamental groups being the automorphism groups of the objects of the said groupoid. The set $\pi_1(S; a', a)$ may be called the *set of classes of paths from a to a'* . These classes therefore compose in the obvious way. Finally, one can define a pro-group Π_1^S of $\mathcal{C}$, which one may call the *fundamental pro-group of S* or *local system of fundamental groups on S* , defined up to unique isomorphism by the condition that one have an isomorphism, functorial in the geometric point a of S :

$$F_a(\Pi_1^S) = \pi_1(S; a)$$

(cf. remark 5.10). In particular, if s is an ordinary point of S , the fiber of G at s is a pro-group over $k(s)$, a projective limit of finite étale groups over $k(s)$; one could call this pro-group *the fundamental group of S at the ordinary point s of S* , and denote it $\pi_1(S, s)$. By definition, these points with values in an algebraically closed extension Ω of $k(s)$ are the elements of $\pi_1(S; a)$, where a is the geometric point of S defined by the said extension. In particular, (taking for S the spectrum of a field) to every field k there is associated canonically and functorially a pro-group over k , which one could denote $\pi_1(k)$, a projective limit of finite étale groups over k , and whose points in an algebraically closed extension Ω of k are identified with the elements of the topological Galois group of $\bar{k}/k$, where $\bar{k}$ is the Galois closure of k in Ω (cf. 8.1). This group $\pi_1(k)$ does not yet seem to have attracted the attention of algebraists.

Now let

$$f: S' \longrightarrow S$$

be a morphism from one connected locally noetherian prescheme into another, let a' be a geometric point of S' and let $a = f(a')$ be its direct image in S . Then the “inverse image” functor induces a functor from the category $\mathcal{C}(S)$ of étale coverings of S into the category $\mathcal{C}(S')$ of étale coverings of S' :

$$f^\bullet: \mathcal{C}(S) \longrightarrow \mathcal{C}(S')$$

One has moreover an isomorphism of functors

$$F_a \cong F_{a'} \circ f^\bullet,$$

so that $f^\bullet$ is an *exact* functor, to which the results of no. 6 apply. In particular one has a canonical homomorphism

$$u = \pi_1(f; a') : \pi_1(S', a') \longrightarrow \pi_1(S; a) \quad (a' = f(a))$$

which makes it possible to reconstruct the inverse image functor, as an operation of restriction of the groups of operators. The properties of the functor $f^\bullet$ are therefore expressed in a simple way by the properties of the associated homomorphism of groups, as was made explicit in no. 6. If in particular S' is an étale covering of S , then the homomorphism u is an isomorphism of $\pi_1(S', a')$ onto the open subgroup of $\pi_1(S, a)$ that defines the pointed connected étale covering S' of S (i.e. the stabilizer U of $a' \in F_a(S')$ in $\pi_1(S; A)$).

If one wishes to interpret the homomorphisms $\pi_1(f; a')$ for a variable geometric point a' , one must, in accordance with what was said in no. 6, consider a homomorphism

$$\Pi_1(f) : \Pi_1^{S'} \longrightarrow f^\bullet(\Pi_1^S)$$

of pro-groups over S , and take the corresponding homomorphism for the geometric fibers.

8. The case of a normal base prescheme

PROPOSITION 8.1. *Let S be the spectrum of a field k , and let Ω be an algebraically closed extension of k , defining a geometric point a of S with values in Ω . Let $\bar{k}$ be the separable closure of k in Ω . Then there exists a canonical isomorphism of $\pi_1(S, a)$ onto the topological Galois group of $\bar{k}/k$.*

Let k' be the algebraic closure of k in Ω ; there therefore corresponds to it a geometric point b of S , with values in k' . The natural homomorphism of functors $F_b \rightarrow F_a$ is evidently an isomorphism, since a k -homomorphism of a finite separable extension of k into Ω necessarily takes its values in $\bar{k}$ and a fortiori in k' . On the other hand, the group π' of k -automorphisms of k'/k operates in the obvious way on F_b , whence a homomorphism $\pi' \rightarrow \text{Aut}(F_b) \xrightarrow{\sim} \text{Aut}(F_a) = \pi_1(S; a)$. On the other hand, it is well known that the natural homomorphism from π' into the group π of automorphisms of $\bar{k}/k$ is an isomorphism. One thus obtains a canonical homomorphism $\pi \rightarrow \pi_1(S; a)$; it remains to show that it is an isomorphism. Indeed, this homomorphism is injective, since an element of the kernel is an automorphism of $\bar{k}/k$ which induces the identity on every finite separable subextension, hence is trivial. This homomorphism is surjective, since if X is a *connected* étale covering of S , hence defined by a finite separable extension L/k , then π is transitive on the set of k -homomorphisms of L into k' , as is well known.

PROPOSITION 8.2. *Let S be a connected, locally noetherian and normal prescheme, $K = k(s)$ its function field = the residue field at its generic point s , and Ω an algebraically closed extension of K , defining a geometric point a' of $S' = \text{Spec}(K)$ and a geometric point a of S . Then the homomorphism $\pi_1(S'; a') \rightarrow \pi_1(S; a)$ is surjective. When one identifies the first group with the Galois group of the separable closure $\bar{K}$ of K in Ω (cf. 8.1), then the kernel of the preceding homomorphism corresponds by Galois theory to the subextension of $\bar{K}/K$ composed of the finite extensions of K in Ω which are unramified over S .*

The first assertion means that the inverse image on S' of a connected étale covering X of S is connected, i.e. that X is integral; this is none other than (I 10.1). The kernel of the preceding homomorphism is then interpreted as consisting of the automorphisms of $\bar{K}/K$ which induce the identity on the sets $F_a(X)$, where one may assume the étale covering X of S connected. But this means that this automorphism induces the identity on the finite subextensions of $\bar{K}/K$ which are unramified over S , which proves the last assertion.

REMARK. Thanks to this interpretation of the fundamental group of the normal prescheme S in terms of the usual Galois theory, the definition had been known in this case for a long time.

9. The case of non-connected preschemes: multi-Galois categories

Let S be a locally noetherian prescheme, and let $(S_i)_{i \in I}$ be its connected components. Then the category $\mathcal{C}(S)$ of étale coverings of S is equivalent to the product category of the $\mathcal{C}(S_i)$, which are interpreted in terms of the fundamental groups of the S_i , once a geometric point has been chosen in each S_i . In the application of descent theory for étale morphisms, it is sometimes inconvenient to choose, for every S_i , a geometric point of S_i . It is then more convenient to resort to the natural generalization of 5.8 in order to interpret $\mathcal{C}(S)$ as a category of functors on the groupoid of geometric points of S , considered as the sum of the groupoids corresponding to the connected components of S ; the functors in question are the functors with values in the category of finite sets satisfying the continuity property analogous to the one invoked in 5.8. In practice, one will have a family $(a_t)_{t \in E}$ of geometric points of S such that every connected component S_i of S contains at least one of them, and one can then, as in 5.10, replace the groupoid of all geometric points of S by the analogous groupoid whose underlying set is E . Of course, these considerations ought to fit into general definitions concerning the categories which are equivalent to product categories of categories

of the form $\mathcal{C}(\pi)$, and which one may call *multi-Galois categories*. We shall leave the details to the reader.